



**VIGNAN'S INSTITUTE OF ENGINEERING FOR WOMEN (A)**

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## DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

# FORECASTING POLLEN-BASED AIR QUALITY USING NEURAL NETWORKS AND ENSEMBLE MODELS

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# Contents

1. Abstract
2. Introduction
3. Literature Survey
4. Problems with Existing System
5. Problem Definition
6. Proposed system
7. Architecture of Proposed System
8. Modules in Proposed System
9. Experimental Analysis(Result)
10. Comparison with Existing System
11. Conclusion
12. Future Scope



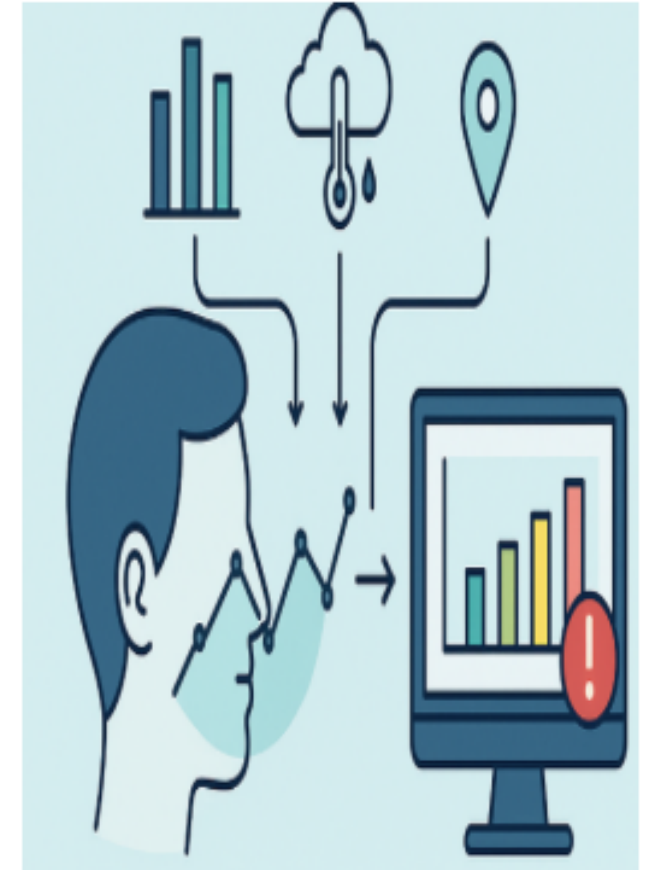
# Abstract

POLLULENS is an innovative AI-powered solution that transforms how we monitor, analyze, and predict air quality. By harnessing the capabilities of machine learning and deep learning technologies, POLLULENS processes vast environmental datasets—ranging from historical pollutant levels and meteorological data to geospatial insights—to deliver highly accurate air quality forecasts. It excels in identifying hidden pollution patterns, detecting unusual environmental changes, and locating emission sources that might otherwise remain unnoticed. Incorporating advanced methodologies like ensemble learning, time-series forecasting, and neural networks, POLLULENS ensures precise and reliable predictions. The platform converts complex environmental data into intuitive dashboards and visualizations, offering accessible and actionable insights for governments, researchers, and the general public. Real-time alerts during critical pollution events allow for swift responses, enhancing public health protection. Beyond forecasting, POLLULENS supports long-term sustainability goals. It assists in urban planning by recommending strategic actions based on predictive analytics, encouraging smarter and greener development. By shifting the focus from reactive measures to proactive, data-driven decision-making, POLLULENS sets a new standard in environmental monitoring. In essence, POLLULENS is more than a monitoring tool—it is a comprehensive AI-based system designed to anticipate air quality risks, safeguard public well-being, and pave the way for a cleaner, healthier future.



# Introduction

POLLULENS is an AI-powered system that changes the way we monitor and predict air quality. It uses machine learning and deep learning to study large amounts of data—like past pollution levels, weather conditions, and location-based information—to give accurate air quality forecasts. It helps find hidden pollution patterns, detect unusual changes, and identify sources of pollution that are hard to spot. With advanced techniques like ensemble learning and time-series forecasting, POLLULENS provides reliable results. The system turns complex data into easy-to-read dashboards and visuals, making it useful for governments, researchers, and the public. It also sends real-time alerts during high pollution events, helping protect public health. Beyond just forecasting, POLLULENS supports smart city planning by giving useful suggestions based on data. It helps shift from reacting to pollution problems to preventing them, aiming for cleaner, healthier, and more sustainable cities. In short, POLLULENS is more than just a monitoring tool—it's a complete AI solution to predict risks, protect health, and promote a better environment.







# Literature Survey

| S.NO | YEAR | AUTHORS                     | TITLE                                                                                    | OUTCOMES                                                                                                                                                              |
|------|------|-----------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | 2025 | Azeraf et al.               | Real-time Pollutant Identification through Optical PM Micro-Sensor                       | Achieved accurate pollutant classification in real time using only optical micro-sensor data. Shows potential for actionable urban air planning support.              |
| 2.   | 2025 | Rathbone et al.             | Using low-cost sensors to assess common air pollution sources across multiple residences | Modeled indoor/outdoor PM variations with very high $R^2$ ( $\geq 0.983$ ); demonstrated spatial and inter-household exposure differences; WHO guideline exceedances. |
| 3.   | 2023 | MIT Senseable city lab team | City Scanner/Flatburn mobile pollution detector                                          | Provides low-cost, widely deployable mobile air quality monitoring; democratizes community-scale data collection                                                      |
| 4.   | 2019 | Concas et al.               | Low-Cost Outdoor Air Quality Monitoring and Sensor Calibration: A Survey                 | Highlights challenges (cross-sensitivities, drift, environmental interference) and shows efficacy of ML-based calibration in improving accuracy.                      |



# Problems with Existing System

- **Lack of Pollen-Specific Forecasting :** Most existing systems do not consider pollen count, even though it significantly impacts individuals with allergies and respiratory illnesses.
- **Low Prediction Accuracy :** Traditional models often use basic statistical or linear regression techniques, which struggle to handle the complex, non-linear nature of environment data, leading to less accurate forecasts.
- **Limited Real-Time Integration :** Many systems are not connected to real-time APIs or live data sources, resulting in delayed or outdated information that limits timely decision-making.
- **No Personalization :** Current platforms lack user-specific recommendations or health alerts tailored to individuals with conditions like asthma or allergic rhinitis.
- **Lack of Integration with Health Systems :** Existing solutions rarely work in collaboration with health agencies to issue public health warnings or suggest preventive actions.



# Problem Definition

Air quality monitoring systems have become increasingly important due to rising pollution levels and their adverse effects on public health. However, existing forecasting models primarily focus on general air pollutants and often overlook pollen-induced pollution, which significantly affects individuals with respiratory conditions such as asthma and allergic rhinitis. Moreover, traditional systems suffer from limited prediction accuracy, lack of real-time data integration, poor personalization, and restricted geographic coverage. They rarely employ advanced machine learning or deep learning techniques to handle complex environmental patterns, and most lack user-friendly interfaces or alert mechanisms. Therefore, there is a need for a robust, intelligent, and accessible system that not only forecasts overall air quality but also incorporates pollen-related environmental factors using advanced modeling techniques, and delivers real-time, personalized insights to support public health and sustainable living.







# Proposed System

Our project introduces an intelligent, AI-powered system for forecasting air quality with a focus on pollen-based pollution. The system uses a hybrid model that integrates LSTM, Random Forest, and XGBoost to improve prediction accuracy.

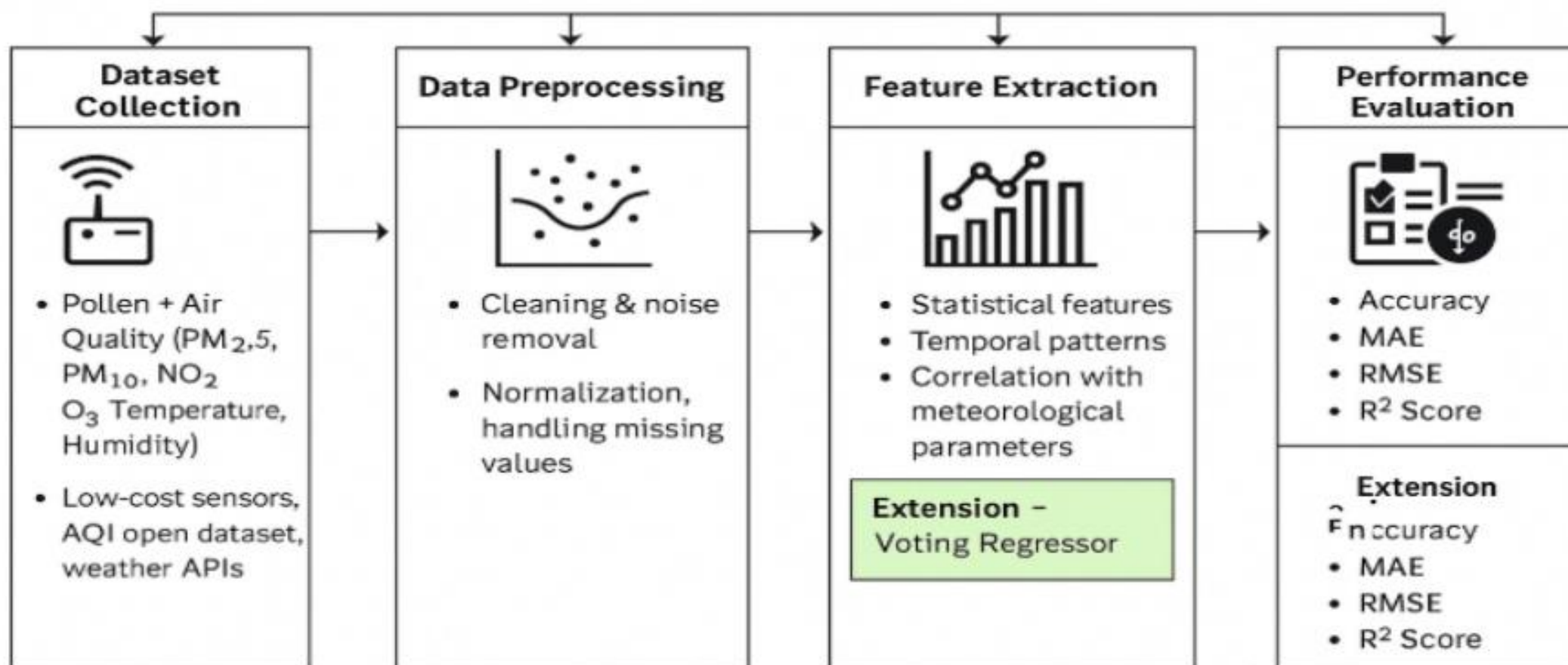
- Real-time Data is fetched from environmental APIs, including pollen count, temperature, humidity, and wind speed.
- Web Dashboard (Streamlit) provides an interactive interface for users to view air quality trends and forecasts.
- Alert System (planned) will deliver health-based alerts tailored to user profiles like asthma or allergy sufferers.
- Advanced Modeling captures both temporal and non-linear patterns in the data for robust AQI predictions.
- Evaluation Metrics such as  $R^2$  (0.94), MAE, and RMSE confirm the accuracy and effectiveness of the model.





# Architecture of proposed System

## Pollulens





# Modules in Proposed System

- **Data Collection and Preprocessing Module :**Collects real-time data from external APIs, including pollen count, temperature, humidity, wind speed, and AQI values. Cleans and normalizes the data, handles missing values, and prepares the dataset for accurate and efficient model training.
- **Feature Extraction Module :** Analyzes environmental variables to identify relevant features that influence AQI and pollen levels. Extracted features such as meteorological parameters and pollen concentration are used to build robust predictive models.
- **Prediction Module :** Implements three models—LSTM, Random Forest, and XGBoost—to predict AQI and pollen concentration. These models capture both temporal and feature-based patterns in the environmental data.
- **Hybrid Modeling Module :** Combines outputs from individual models into a single, optimized hybrid model using ensemble techniques. This improves prediction accuracy and robustness, achieving an  $R^2$  score of 0.94.
- **Evaluation Module :** Evaluates the performance of individual and hybrid models using metrics such as  $R^2$ , MAE, and RMSE. Performs cross-validation to ensure model reliability on unseen data.



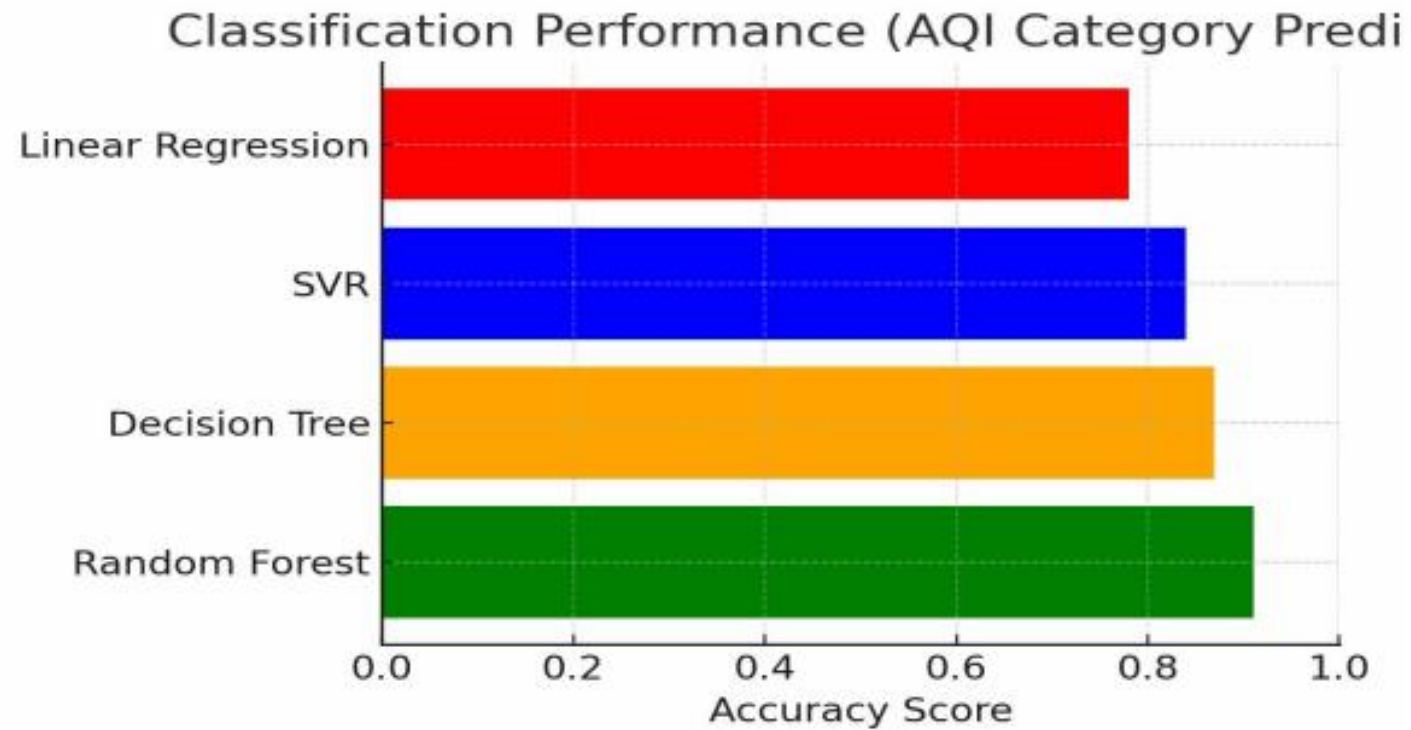
# Experimental Analysis

- By utilizing advanced deep learning models such as Xception, ResNet50, InceptionV3, and VGG19 for feature extraction, the system achieves superior classification performance in handwriting-based detection.
- These models effectively identify intricate motor impairments and handwriting irregularities that are often challenging to detect through conventional methods.
- Among them, Xception-based models have demonstrated remarkable potential, with preliminary results suggesting near-perfect accuracy. This highlights their ability to capture the subtle variations in handwriting patterns that serve as early indicators of Parkinson's disease.
- The detection framework can provide a more comprehensive and holistic assessment of Parkinson's, improving diagnostic precision and offering deeper insights into disease progression. Develop a highly accurate and efficient Parkinson's disease detection system that surpasses traditional diagnostic approaches.



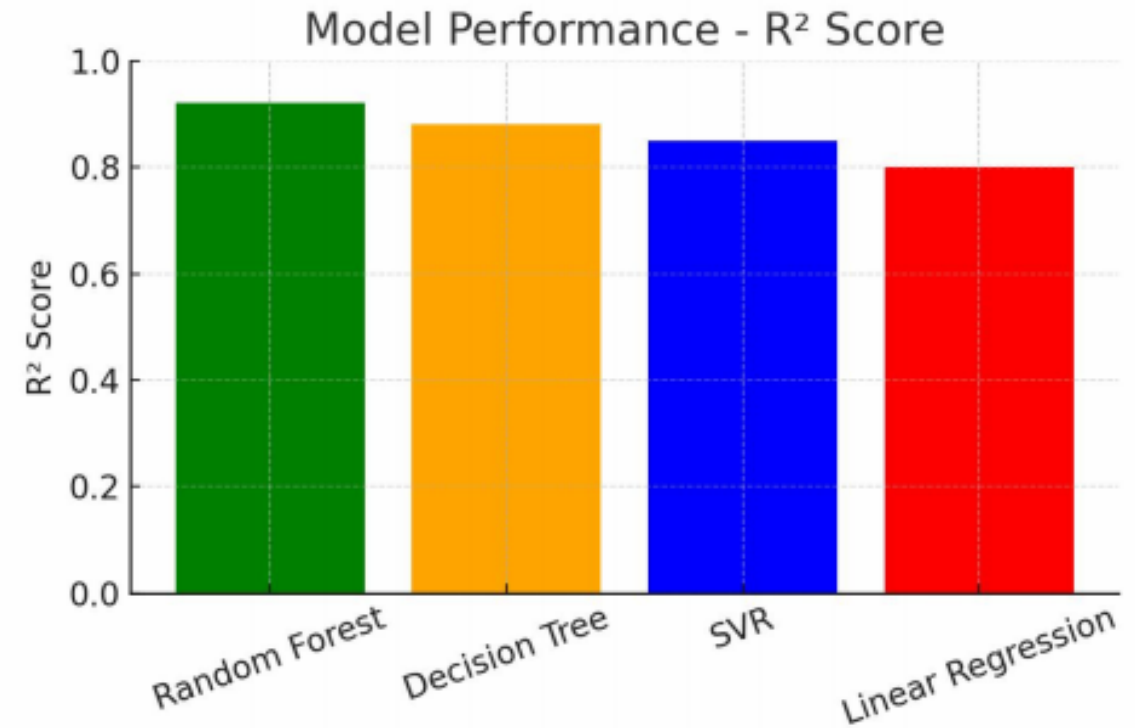
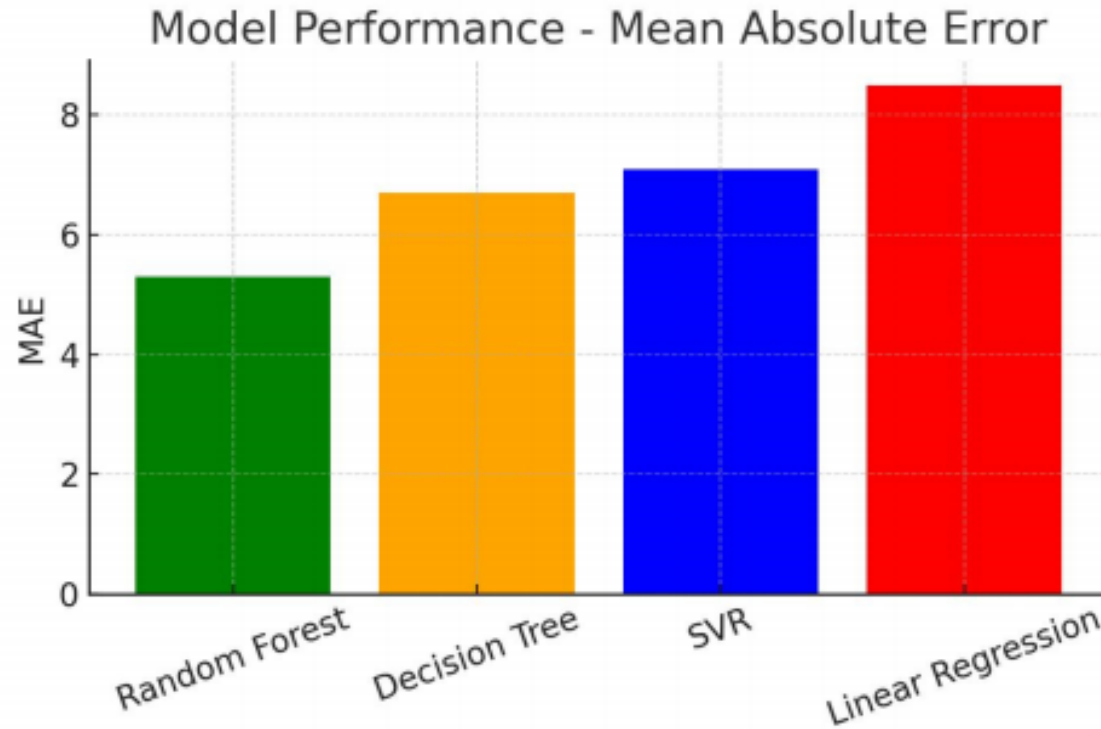


# Experimental Analysis





# Experimental Analysis





# Comparison

| Aspect                   | Existing Systems                                             | Proposed System                                                           |
|--------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------|
| Model Accuracy           | Uses simple models with lower prediction accuracy            | Employs hybrid models (LSTM + RF + XGBoost) with high $R^2$ score (0.94)  |
| Forecasting Capability   | Limited or no future prediction; mostly real-time analysis   | Predictive system with time-series forecasting for AQI & pollen levels    |
| Alert System             | Generic alerts, not personalized                             | Real-time, personalized alerts based on user's sensitivity (e.g., asthma) |
| Environmental Parameters | Ignores environmental conditions like wind, humidity, pollen | Considers temperature, humidity, wind speed, and pollen count             |
| Health Awareness Support | Lacks integration with health impact assessment              | Supports public health protection through alert-based recommendations     |
| Cost                     | High due to manual operations, lack of automation            | Cost-efficient with automation, open-source tools like Streamlit & APIs   |
| Scalability              | Limited to local or regional setups                          | Cost-efficient with automation, open-source tools like Streamlit & APIs   |





# Conclusion

- A hybrid air quality forecasting system was developed using LSTM, Random Forest, and XGBoost models.
- The model achieved high prediction accuracy ( $R^2 = 0.94$ ), proving effective for both AQI and pollen level forecasting.
- Real-time data integration from APIs was implemented, including features like pollen count, humidity, temperature, and wind speed.
- Proper data preprocessing ensured enhanced model performance and reduced noise in predictions.
- A Streamlit-based web dashboard was built for users to interact with the system and view forecasts easily.
- The system provided personalized alerts to raise awareness about air quality risks, especially for sensitive individuals.
- The model was validated using real-world datasets and evaluation metrics like MAE, RMSE, and  $R^2$ , confirming its reliability.
- The solution contributes to public health protection and supports sustainable urban planning.



# Future Scope

- **Geographic Coverage Expansion** : Currently limited to selected regions; future versions can scale to nationwide or global coverage by integrating satellite and international meteorological data sources.
- **Personalized Health Alerts** : Introduce user profiling to provide custom health advisories for people suffering from asthma, allergies, or other pollen-triggered conditions.
- **Environmental Policy Support** : Provide data insights to government or healthcare agencies for planning and public health decision-making.
- **Forecasting Other Environmental Hazards** : Extend the system to predict smog, forest fire smoke, or dust storm intrusions, in addition to pollen.
- **Live Data Stream and Continuous Learning** : Implement real-time learning pipelines to continuously improve model performance as new data (pollutants, weather, pollen) arrives via APIs.
- **Mobile Application Integration** : Develop a mobile app for real-time AQI and pollen alerts, allowing users (especially allergy-prone or asthma patients) to receive location-based warnings instantly.



THANK YOU  
Queries?